

Banach-Space Partial Results for Kadets's Convex-Body Inradius Covering Problem

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Abstract

Kadets proved that if a convex body A in a Hilbert space is covered by convex bodies A_n , then

$$\sum_n r(A_n) \geq r(A),$$

and asked whether the same statement holds in arbitrary Banach spaces. We prove the exact inequality in every real Banach space when all covering bodies A_n are centrally symmetric. We also prove a comparison theorem: if the norm is within distortion D of a Hilbertian norm, then

$$\sum_n r(A_n) \geq \frac{1}{D} r(A).$$

John's ellipsoid theorem consequently gives the universal factor $1/\sqrt{d}$ in dimension d , improving the elementary factor $2/(d+1)$ obtained here by converting covering bodies into containing planks and invoking Ball's Banach-space plank theorem. The unrestricted dimension-free problem remains open.

1 Source question

Kadets's paper [1] proves the inequality in real and complex Hilbert spaces. On PDF page 1, at the end of the Introduction, it states:

An analogous question for general Banach spaces remains open.

Thus the extracted question is whether, for every Banach space X , every closed convex body $A \subset X$, and every sequence of closed convex bodies (A_n) covering A , one necessarily has

$$\sum_{n=1}^{\infty} r_X(A_n) \geq r_X(A).$$

Here is the page-one source passage; the crop retains the full readable page width and the complete open statement.

COVERINGS BY CONVEX BODIES AND INSCRIBED BALLS

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ABSTRACT. Let H be a Hilbert space. For a closed convex body A denote by $r(A)$ the supremum of radii of balls, contained in A . We prove, that $\sum_{n=1}^{\infty} r(A_n) \geq r(A)$ for every covering of a convex closed body $A \subset H$ by a sequence of convex closed bodies A_n , $n \in \mathbb{N}$. It looks like this fact is new even for triangles in a 2-dimensional space.

1. INTRODUCTION

Recall, that by a plank of width w in a Hilbert space one means a set of the form

$$P = \{h \in H : |\langle h - h_0, e \rangle| \leq \frac{w}{2}\},$$

where $\|e\| = 1$. According to T.Bang's theorem [3], if a sequence P_n of planks of widths w_n covers a ball of diameter w , then $\sum w_n \geq w$. (To be more precise this is a particular case of Bang's theorem. The complete statement includes a convex set P of minimal width w instead of a ball).

K. Ball generalized the Bang's theorem to coverings of a ball in a Banach space, for planks defined with help of linear functionals instead of inner product.

Our work is inspired by Bang's and Ball's theorems. The idea was to find a "symmetric" generalization of this theorems, where the elements of the covering and the covered set are of the same nature, and they are measured "from inside". In this paper we prove such a generalization for sets in a Hilbert space. An analogous question for general Banach spaces remains open. In the proof we use ideas from Ball's exposition of Bang's theorem.

2. THE MAIN RESULT

The same question is still recorded as open for general Banach spaces in the published version of Verreault's survey [5, Section 4.4].

2 Definitions and the plank input

Throughout, X is a real Banach space with closed unit ball B_X . For a closed convex set $C \subset X$ with nonempty interior, its inradius is

$$r_X(C) = \sup\{t \geq 0 : x + tB_X \subset C \text{ for some } x \in X\}.$$

We allow the value $+\infty$. A norm plank is a set

$$P(f; a, b) = \{x \in X : a \leq f(x) \leq b\}, \quad f \in X^*, \quad \|f\| = 1,$$

and its width is $b - a$.

We use Ball's solution of the plank problem for symmetric bodies [2]: if a finite or countable family of planks covers a norm ball $x + qB_X$, then the sum of their widths is at least $2q$. This countable form is the one quoted by Kadets [1, p. 1]. In finite dimensions it also follows from the finite theorem: widen the n -th closed plank by total width $\varepsilon 2^{-n}$. The interiors of the widened planks cover the compact ball, hence have a finite subcover. Apply the finite theorem and let $\varepsilon \downarrow 0$.

3 Proof intuition

The source problem asks for an inside measurement of arbitrary covering bodies, whereas Ball's theorem controls an outside measurement of planks. We bridge the two measurements.

For a centrally symmetric body C , any ball contained off-center has its reflection contained as well; averaging the two balls puts a ball of the same radius at the symmetry center. A supporting direction that nearly realizes this centered inradius traps C in a plank of width nearly $2r_X(C)$. Replacing every A_n by such a plank loses no constant, so Ball gives the exact Kadets inequality.

For a nonsymmetric compact body C in dimension d , the active supporting functionals at an incenter balance: zero lies in their convex hull. By Carathéodory, at most $d+1$ active functionals are needed. One has weight at least $1/(d+1)$, and the balancing identity bounds the opposite support in that direction. The resulting containing plank has width at most $(d+1)r_X(C)$. Applying this to the intersections $A_n \cap U$, where $U \subset A$ is an inscribed norm ball, and then applying Ball yields the factor $2/(d+1)$.

There is also a stronger global comparison. If a Hilbert ellipsoid E satisfies $E \subset B_X \subset DE$, then the two inradii satisfy $r_E(C)/D \leq r_X(C) \leq r_E(C)$. Applying Kadets's Hilbert theorem in the E -norm immediately transfers the exact Hilbert inequality with loss only D . John's theorem supplies $D \leq \sqrt{d}$. These results solve a broad structured subcase and give a universal finite-dimensional approximation, but neither resolves the unrestricted dimension-free question.

4 The exact centrally symmetric covering-body case

For a closed convex set $K \subset X$ containing 0, write its support function in the extended sense as

$$h_K(f) = \sup_{x \in K} f(x) \in \mathbb{R} \cup \{+\infty\}.$$

Lemma 1 (Symmetric bodies fit their inradius plankwise). *Let $C = c + K$ be a closed centrally symmetric convex body in X , where $K = -K$. If $r_X(C) < \infty$, then for every $\eta > 0$ there is a norm plank containing C whose width is at most $2r_X(C) + \eta$.*

Proof. Suppose $x + tB_X \subset K$. By symmetry, $-x + tB_X \subset K$. Convexity and pointwise averaging imply $tB_X \subset K$. Thus every admissible radius can be realized at the symmetry center and

$$r_X(C) = \sup\{t : tB_X \subset K\} = \inf_{\|f\|=1} h_K(f).$$

For the second equality, one implication follows by applying f to $tB_X \subset K$. Conversely, if $t < \inf_{\|f\|=1} h_K(f)$, then $tB_X \subset K$ by the Hahn–Banach separation representation of a closed convex set as the intersection of its supporting half-spaces.

Choose $f \in X^*$, $\|f\| = 1$, with $h_K(f) \leq r_X(C) + \eta/2$. Since $K = -K$, $h_K(-f) = h_K(f)$, and hence C is contained in

$$\{y : f(c) - h_K(f) \leq f(y) \leq f(c) + h_K(f)\}.$$

This plank has width $2h_K(f) \leq 2r_X(C) + \eta$. □

Theorem 1 (Exact inequality for symmetric covering bodies). *Let X be any real Banach space. Suppose a closed convex body $A \subset X$ is covered by a sequence of closed centrally symmetric convex bodies $(A_n)_{n \geq 1}$. Their symmetry centers may be different. Then*

$$\sum_{n=1}^{\infty} r_X(A_n) \geq r_X(A).$$

Proof. If $S := \sum_n r_X(A_n) = +\infty$, there is nothing to prove. Otherwise every summand is finite. Fix $\varepsilon > 0$, choose positive η_n with $\sum_n \eta_n < \varepsilon$, and use Lemma 1 to put each A_n in a plank P_n of width at most $2r_X(A_n) + \eta_n$. The planks cover A .

For every $q < r_X(A)$, some translate of qB_X lies in A , hence is covered by the P_n . The ball plank theorem gives

$$2q \leq \sum_n \text{width}(P_n) \leq 2S + \varepsilon.$$

Letting $\varepsilon \downarrow 0$ and then $q \uparrow r_X(A)$ proves the claim. The same argument covers $r_X(A) = +\infty$, because q may then be arbitrarily large. $\square$

Remark 1. Theorem 1 includes covers by translates of norm balls and by planks. It does not require the covered body A itself to be symmetric.

5 A Hilbertian-distortion bound

For the unit ball E of a Hilbertian norm on X , write $r_E(C)$ for the corresponding inradius.

Theorem 2 (Transfer from a Hilbertian norm). *Suppose that X admits a Hilbertian norm whose closed unit ball E satisfies*

$$E \subset B_X \subset DE$$

for some $D \geq 1$. If a closed convex body $A \subset X$ is covered by a sequence of closed convex bodies (A_n) , then

$$\sum_{n=1}^{\infty} r_X(A_n) \geq \frac{1}{D} r_X(A).$$

Proof. For every closed convex set C , the inclusions of the unit balls give

$$\frac{1}{D} r_E(C) \leq r_X(C) \leq r_E(C). \quad (2)$$

Indeed, an E -ball of radius t contains an X -ball of radius t/D , while every X -ball of radius t contains the corresponding E -ball. Kadets's theorem applied in the Hilbertian norm gives

$$\sum_n r_E(A_n) \geq r_E(A).$$

Combining this with (2) yields

$$\sum_n r_X(A_n) \geq \frac{1}{D} \sum_n r_E(A_n) \geq \frac{1}{D} r_E(A) \geq \frac{1}{D} r_X(A),$$

including the extended-value cases. $\square$

Corollary 1 (Universal finite-dimensional bound). *If X is a d -dimensional real normed space and (A_n) covers A , then*

$$\sum_{n=1}^{\infty} r_X(A_n) \geq \frac{1}{\sqrt{d}} r_X(A).$$

Proof. John's ellipsoid theorem for the centrally symmetric unit ball B_X provides an ellipsoid E with $E \subset B_X \subset \sqrt{d} E$ [3]. Apply Theorem 2 with $D = \sqrt{d}$. $\square$

6 An elementary containing-plank bound

Lemma 2 (A $(d+1)$ -inradius containing plank). *Let X be a d -dimensional normed space, $d \geq 1$, and let $C \subset X$ be a nonempty compact convex set. Then C is contained in a norm plank of width at most*

$$(d+1)r_X(C).$$

Here $r_X(C) = 0$ when C has empty interior.

Proof. If C is lower-dimensional, its affine hull is contained in a hyperplane, which is a plank of width $0 = (d+1)r_X(C)$. Assume henceforth that C has nonempty interior. Compactness gives an incenter c ; translating, take $c = 0$, and set $\rho = r_X(C) > 0$. For $f \in S_{X^*}$, put $h(f) = h_C(f)$. Then

$$\rho = \min_{f \in S_{X^*}} h(f).$$

Let $F = \{f \in S_{X^*} : h(f) = \rho\}$ be the nonempty compact set of active supporting functionals.

We first show that

$$0 \in \text{conv } F. \tag{1}$$

If not, strict separation in the finite-dimensional dual gives a vector $v \in X$ and $\alpha > 0$ such that $f(v) \geq \alpha$ for every $f \in F$. By continuity, $f(v) \geq \alpha/2$ on a neighborhood N of F in S_{X^*} . On the compact complement there is $\delta > 0$ such that $h(f) \geq \rho + \delta$. For all sufficiently small $t > 0$, therefore,

$$h(f) - f(-tv) = h(f) + tf(v) > \rho \quad (f \in S_{X^*}).$$

This says that a ball of radius strictly larger than ρ , centered at $-tv$, is contained in C , contradicting maximality of the inball. Thus (1) holds.

Carathéodory's theorem now gives $f_1, \dots, f_m \in F$, positive numbers $\lambda_1, \dots, \lambda_m$, and $m \leq d+1$ such that

$$\sum_{j=1}^m \lambda_j = 1, \quad \sum_{j=1}^m \lambda_j f_j = 0.$$

Choose i with $\lambda_i \geq 1/m \geq 1/(d+1)$. For every $x \in C$, all $f_j(x) \leq h(f_j) = \rho$, and the balancing identity gives

$$-f_i(x) = \frac{1}{\lambda_i} \sum_{j \neq i} \lambda_j f_j(x) \leq \rho \frac{1 - \lambda_i}{\lambda_i}.$$

Together with $f_i(x) \leq \rho$, this places C in the plank

$$-\rho \frac{1 - \lambda_i}{\lambda_i} \leq f_i(x) \leq \rho,$$

whose width is

$$\frac{\rho}{\lambda_i} \leq (d+1)\rho.$$

□

Theorem 3 (Elementary dimension-dependent inequality). *Let X be a d -dimensional real normed space. If a closed convex body $A \subset X$ is covered by a sequence of closed convex bodies (A_n) , then*

$$\sum_{n=1}^{\infty} r_X(A_n) \geq \frac{2}{d+1} r_X(A).$$

Proof. Fix $q < r_X(A)$ and choose a closed norm ball $U = x_0 + qB_X \subset A$. Put $C_n = A_n \cap U$. Each nonempty C_n is compact and convex, the C_n 's cover U , and $r_X(C_n) \leq r_X(A_n)$. By Lemma 2, choose a plank $P_n \supset C_n$ with

$$\text{width}(P_n) \leq (d+1)r_X(C_n).$$

The P_n 's cover U . The countable ball plank theorem yields

$$2q \leq \sum_n \text{width}(P_n) \leq (d+1) \sum_n r_X(C_n) \leq (d+1) \sum_n r_X(A_n).$$

Let $q \uparrow r_X(A)$. If $r_X(A) = +\infty$, the same inequality for arbitrarily large q forces the sum on the right to be infinite. $\square$

Corollary 2. *The distortion bound gives guaranteed factors $1, 1/\sqrt{2}, 1/\sqrt{3}$ in dimensions 1, 2, 3. The elementary bound is never stronger, since $2/(d+1) \leq 1/\sqrt{d}$, with equality only when $d = 1$.*

7 Verification and bounded computational check

The proof has five independently checkable obligations.

1. Central symmetry really centers the inradius: reflecting and averaging an arbitrary inscribed ball preserves its radius.
2. The active functionals at an incenter must balance. The proof above includes the compactness step needed to turn strict separation into a uniform improvement of every support slack.
3. The Carathéodory calculation is exact: the selected coefficient $\lambda_i \geq 1/(d+1)$ produces plank width ρ/λ_i , not twice that quantity.
4. Intersecting with U is used only in the nonsymmetric finite-dimensional proof. It is deliberately avoided in the symmetric proof, since intersections of symmetric bodies with different centers need not be symmetric.
5. The norm comparison has the correct direction: from $E \subset B_X \subset DE$ one gets $r_E(C)/D \leq r_X(C) \leq r_E(C)$. Thus only one factor D , rather than D^2 , is lost when Kadets's Hilbert theorem is transferred.

Before the plank argument was identified, the script `code/check_cube_triangulations.py` used linear programming to compute the largest axis-parallel cube in each cell of two standard tetrahedral partitions of $[0, 1]^3$ in the ℓ_∞^3 norm. The five-tetrahedron partition has total inradius $5/6$, and the six-cell Freudenthal partition has total inradius 1, compared with ambient cube inradius $1/2$. Thus neither is a counterexample. This bounded computation is not used in either proof.

8 Limitations and novelty check

Theorem 1 assumes symmetry of every covering body, not only of A . Corollary 1 loses the factor $1/\sqrt{d}$, so it does not settle the exact question for any nonsymmetric finite-dimensional normed space of dimension at least two. Theorem 3 gives the weaker factor $2/(d+1)$ by an independent elementary route. Kadets's sharper constant 1 remains available in Hilbert spaces [1]. Akopyan

and Karasev prove exact relative-inradius results for arbitrary planar partitions and for inductive partitions in higher dimension [4]; a general cover need not be a partition.

Before promotion, the run’s registry and solution indexes were searched for arXiv:math/0312133 and the core phrases *convex-body cover inradius*, *centrally symmetric covering bodies*, *Hilbertian distortion*, $1/\sqrt{d}$, $2/(d+1)$, and *containing plank $d+1$ inradius*. A bounded web/arXiv search used the same phrases and located Kadets [1], Ball [2], Akopyan–Karasev [4], and Verreault’s current survey [5], but no exact match for either theorem above. Novelty confidence is therefore moderate: these short syntheses of classical tools may be folklore, and no priority claim is made.

9 Human-review recommendation

Classify this as a *likely valid strengthened partial result*. A verifier should focus on the norm-comparison directions in Theorem 2, Lemmas 1 and 2, and the countable-plank passage. The packet should not be labeled a full solution unless the factor $1/\sqrt{d}$ is raised to 1 without symmetry and with no dimension dependence.

References

- [1] V. Kadets, *Coverings by convex bodies and inscribed balls*, Proc. Amer. Math. Soc. **133** (2005), 1491–1495; arXiv:math/0312133.
- [2] K. Ball, *The plank problem for symmetric bodies*, Invent. Math. **104** (1991), 535–543; arXiv:math/9201218.
- [3] F. John, *Extremum problems with inequalities as subsidiary conditions*, in *Studies and Essays Presented to R. Courant on His 60th Birthday*, Interscience, 1948, pp. 187–204.
- [4] A. Akopyan and R. Karasev, *Kadets type theorems for partitions of a convex body*, arXiv:1106.5635.
- [5] W. Verreault, *Plank theorems and their applications: a survey*, Bull. Lond. Math. Soc. (2026), DOI: 10.1112/blms.70230; arXiv:2203.05540v2.